B1.2 Proteins

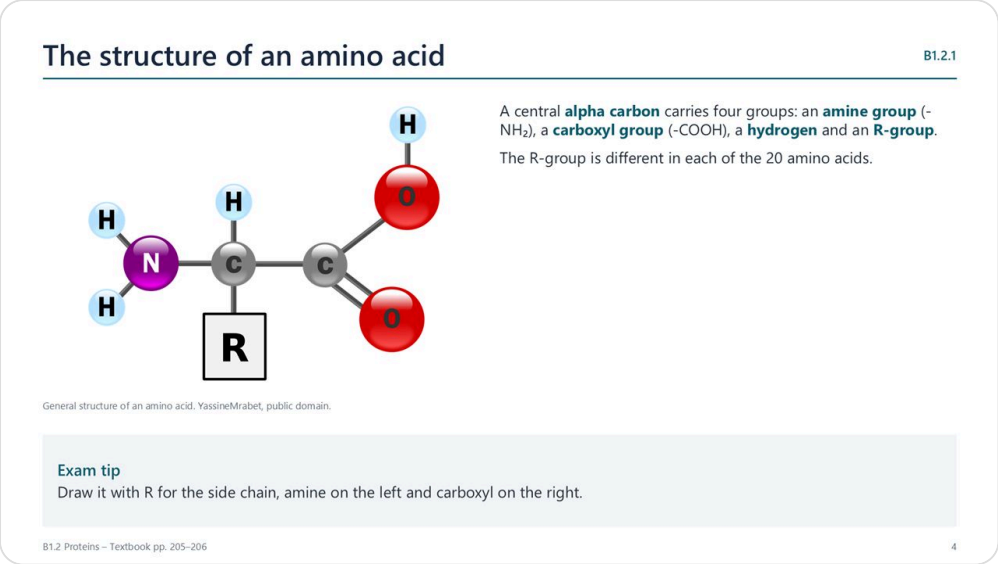
Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b1-2.html

Written for this site from the class slides. The chapter is on pages 205–210 of the course textbook.
Image credits are on the last slide of the slides for this subtopic.

B1.2.1 The structure of an amino acid

pp. 205–210

Four groups are attached to the **alpha carbon**: an **amine group**, a **carboxyl group**, a **hydrogen atom** and an **R-group**. Only the R-group differs between amino acids.



B1.2.2 Condensation forms dipeptides and longer chains

pp. 205–210

amino acid + amino acid → dipeptide + water. The -OH from a carboxyl group and an -H from an amine group form water, and a **peptide bond** links the two. Repeating this adds amino acids to make a polypeptide.

Watch out: a protein contains one or more polypeptides, folded into a working shape.

B1.2.3 Dietary requirements

pp. 205–210

Essential amino acids cannot be made, and must come from food. **Non-essential** ones can be made from other amino acids. Vegan diets need attention, to ensure that all the essential amino acids are included.

B1.2.4 The limitless variety of peptide chains

pp. 205–210

There are 20 amino acids in the genetic code. They can be in any order, and chains can have from a few to thousands of amino acids. A chain of 3 can have 8000 sequences, and one of 6 can have 64 million.

Protein	Function
Haemoglobin	Carries oxygen in red blood cells
Immunoglobulin	Antibody that binds to antigens
Rhodopsin	Pigment in the retina, absorbs light
Collagen	Rope-like strength in skin and tendons
Insulin	Hormone that lowers blood glucose

B1.2.5 Effects of pH and temperature

pp. 205–210

The shape of a protein is held by weak bonds and interactions between R-groups. **Heat** makes molecules vibrate, and this breaks those bonds. A change in **pH** alters charges on R-groups, and this breaks attractions. Either can cause **denaturation**: the shape changes, usually permanently, and the protein loses its function.

Watch out: denaturation does not break peptide bonds, so the sequence stays the same.

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